DEPI · DATA ANALYSIS TRACK · FINAL PRESENTATION

RailSmart

UK Railway Intelligence Platform — Presentation Running Order

FORMAT
20 min presentation + 10–15 min discussion

STRUCTURE
4 acts + close · ~23 slides · 2 demos

SCORED AGAINST
DEPI Data Analysis rubric

HEADLINE NUMBER
£133,568 unclaimed compensation

DATASET
31,653 journeys · Jan–Apr 2024 · 65 routes

TEAM
Eissa · Elgohary · Yasser · Nasser · Shaaban

Every figure in this document is reconciled to **cleaned_data/railway_cleaned.csv** and reproducible from the notebook. Minute budgets are cumulative and assume rehearsal to time; the three slides marked **CUT FIRST** are the overflow buffer.

The arc at a glance

Problem → data → analysis → prediction → the built product → forward look. The five tool sections from the first draft collapse into one "four surfaces" slide, freeing five minutes for the analysis the rubric scores.

ACT 0	ACT I	ACT II	ACT III	ACT IV	CLOSE
Frame	**The Data**	**The Analysis**	**Prediction**	**Delivery**	**Land it**
2:30 · 3 slides	4:00 · 4 slides	6:00 · 6 slides	4:00 · 4 slides	4:30 · 3 + demo	3:00 · 3 slides

ACT 0 Frame

2:30 · 3 slides

1	**Title** 0:20	RailSmart · one-line value proposition · five names · DEPI. Keep it to twenty seconds.
2	**The Problem** 1:10	The £133,568 slide. Three stacked facts: 13.2% of journeys disrupted → 73.2% of entitled passengers never claim → £133,568 unclaimed in four months. No adjectives, no stock photos.
3	**Solution + Architecture** 1:00	One diagram: **railway.csv** → **cleaning** → **railway_cleaned.csv** → **four surfaces** (Tableau / Power BI / Web / Mobile) + three personas. This single slide replaces the old "Solution" and "Implementation" sections.

ACT I The Data

4:00 · 4 slides

Rubric: weeks 1–3, 8 (understanding, cleaning, feature engineering).

4	**Dataset & data model** 1:00	Provenance, 31,653 rows, 19 raw columns, 1 Jan – 30 Apr 2024 (121 days), data-dictionary excerpt. Say the date range aloud — it bounds every claim.
5	**Data quality: what was broken** 1:30	The **credibility slide**. Four defects — the `'None'`→NaN trap; colliding delay labels; nulls correctly left as null; and the **fix that broke the next chart** (`.notna()` flagged all 31,653 as railcard holders → £nan). "We caught our own regression" is the most senior thing you can say.
6	**Feature engineering** 1:00	19 → 42 columns. Name the six that matter: Delay_Minutes, Days_In_Advance, Time_Of_Day, Route, Is_Weekend, Price_Per_Minute — and *why* each was derived.
7	**SQL cleaning pipeline** · CUT FIRST 0:30	sql/Cleaning & Analysis.sql, 362 lines. Shows tool breadth. Drop it first if over time.

ACT II The Analysis

6:00 · 6 slides

Rubric: weeks 4–7. This is where the marks are — do not rush it.

8	**Punctuality baseline** 0:45	86.8% on time · 7.2% delayed · 5.9% cancelled. Frame the 13.2% disruption rate as the number that matters.
9	**Where the money leaks** 1:15	The £133,568 unclaimed, broken out: cancelled claim at 30.4%, delayed at 23.8%; online 28.6% vs station 25.8%. No on-time journey ever claimed — the entitlement lives entirely inside the 13.2%.
10	**Why trains are late** 1:00	**Weather is #1** (32.9%), then Signal 23.3%, Staffing 19.4%, Technical 16.9%, Traffic 7.5%. Not signal failure — correct the old README claim before anyone reads it.
11	**Route reliability** 1:00	Worst routes by on-time rate. Edinburgh Waverley → London Kings Cross at 0% across 51 journeys. Show the ≥30-journey threshold — a grader will ask.
12	**The Urgency Tax** 1:00	38.3% premium for booking within 2 days. State the caveat yourself: median lead time is 1 day, max 28, so the "urgent" bucket holds most of the data. Own it before they find it.
13	**Business recommendations** 1:00	**The most important slide, and it doesn't exist yet.** Three actions, each with a £ figure: (1) auto-claim → recover £133,568 / 4mo; (2) weather-linked staffing on the 5 worst routes; (3) nudge station-channel buyers. This is the judgement the rubric tests — only you can write it.

ACT III Prediction

4:00 · 4 slides

Rubric: weeks 9–10 (forecasting + models).

14	**Forecasting** 1:15	SARIMA vs two naive benchmarks. **The honest result: SARIMA can't beat a flat mean** (rides 18.2% vs 18.4% MAPE; revenue 21.6% vs 20.5% — baseline wins). Demand is stationary. Present that as the finding, then give the 30-day projection with its 95% interval (7,927 rides; £183,407).
15	**Predictive models** 1:00	Fare: MAE £2.75, R^2 0.963 — 79% of signal is route, 5% is lead time. Delay risk: AUC 0.979, recall 0.657. Claim propensity: AUC 0.860, recall 0.737.
16	**Honest model evaluation** 1:15	**This slide wins your Q&A.** Baseline vs model, precision/recall/F1, confusion matrices — plus the delay-vs-cancellation table (pooling drops AUC 0.98→0.84). Present your own baseline before the panel finds it.
17	**AI vision — YOLOv8** 0:30	15-second clip, live person count on the platform. Label it clearly as an extension beyond the tabular rubric, not one of your models.

ACT IV Delivery

4:30 · 3 slides + demo

18	**Four surfaces, one dataset** 1:00	2×2 screenshot grid: Tableau · Power BI · Web · Mobile. One arrow from railway_cleaned.csv to all four. Make the architectural claim once — this replaces four of the original sections.
19	**LIVE DEMO — web** 2:30	Three role views. QR code on screen throughout. **Rehearse to the second** — an overrun eats your conclusion. Keep the recorded video cued as a fallback.
20	**Mobile** 1:00	Recorded clip (or live via `expo start --tunnel`, which works off-network). Say "no QR — Expo Go needs the same network" before anyone asks. Owning the constraint reads as competence.

CLOSE Land it

3:00 · 3 slides

21	**Limitations & next steps** 1:00	121 days of data, single dataset, no live API, models are baselines. Honesty is scored — and it pre-empts half the panel's questions.
22	**Future Work — Egypt** 1:15	Three honest columns: what transfers (pipeline, models, dashboards), what doesn't (UK Delay Repay is statutory — Egypt has no equivalent, so the refund feature becomes a policy proposal), what you'd need (12 months of ticketing with scheduled + actual times, one corridor first). **No unsourced Egyptian ridership figures.**
23	**Conclusion + Q&A holding slide** 0:45	Landing line, the explicit ask, repo QR, team names. Leave this slide up for the entire discussion.

Then: 6–10 backup slides (confusion matrices, baseline table, feature importances, full data dictionary, cleaning log, SQL). You have 10–15 minutes of discussion — longer than most acts. Cumulative: 20:00. Buffer comes from slides 7, 17, 20 in that order.

Changes from the original draft: "Solution" + "Implementation" merge into slide 3 · the five tool sections (Tableau / Power BI / Web / Mobile / AI) collapse into slide 18 + two demos · Python expands into three acts (Data / Analysis / Prediction) · the three tabular models move to Act III, YOLOv8 stays with the demo · added: Data Quality, Business Recommendations, Forecasting, Honest Evaluation, Limitations, and backup slides.

Appendix — verified figures to quote

Every number reconciled to cleaned_data/railway_cleaned.csv (36/36 automated checks pass). Safe to put on a slide.

Figure	Value	Figure	Value
Journeys	**31,653**	Columns	**19 → 42**
Journey range	**1 Jan–30 Apr 24**	Routes	**65**
Total revenue	**£741,921**	Mean / median fare	**£23.44 / £11**
On time	**86.8%**	Disrupted	**13.2%**
Delayed / cancelled	**7.2% / 5.9%**	Entitled to refund	**4,172**
Never claimed	**3,054 (73.2%)**	Unclaimed value	**£133,568**
Refunds paid	**£38,702**	Unclaimed / revenue	**18.0%**
Top delay cause	**Weather 32.9%**	Worst hour	**08:00 · 33.4%**
Worst route	**0% on-time**	Top revenue route	**£183,193**
Railcard adoption	**33.9%**	Railcard saving	**42.9% less**
Booking lead (median)	**1 day**	Max lead time	**28 days**
Urgency tax	**38.3%**	Advance vs Anytime	**55% cheaper**
Fare model	**MAE £2.75 · R^2 .96**	Delay model	**AUC .979 · rec .66**
Claim propensity	**AUC .860 · rec .74**	Cancellation model	**AUC .73 · not usable**
Forecast 30-day rides	**7,927**	Forecast 30-day rev	**£183,407**

The questions you will be asked — and the answers

Rehearse a 20-second answer to each. Assign an owner per question.

"96% accurate — what if it just always says on-time?"
→ "92.8%. So accuracy is the wrong metric; we quote AUC 0.979, recall 0.657."

"What's the recall on delayed journeys?"
→ 0.657 at 0.794 precision — we catch two thirds of real delays.

"Is a cancelled train counted as delayed?"
→ "No, deliberately. Pooling them drops AUC 0.98→0.84; cancellation isn't predictable (AUC 0.73). We ship delay as a prediction, cancellation as a route statistic."

"Predicting refunds from delay status — isn't that circular?"
→ "It was. We retrained on the 4,172 entitled only — claim propensity, AUC 0.860, recall 0.737. The naive version stays in as the counter-example."

"Median lead time is 1 day — how recommend booking 14–21 days ahead?"
→ "We don't. Nothing in the data is booked past 28 days; that claim was removed."

"121 days — how can you forecast a month?"
→ "Carefully — and the naive baseline wins. The series is stationary; we forecast the mean with a 95% interval and say so."

"Your railcard chart shows £nan."
→ "It did — .notna() flagged every passenger a holder after we filled blanks with a string. We caught it; real adoption is 33.9%."

"Where's the Tableau dashboard the plan promised?"
→ Build it — docs/TABLEAU_BUILD.md has the order; extracts are generated.